NARAYANA
IIT - JEE ACADEMY - INDIA

Sec: SR.IIT_STAR CO_SC (MODEL-A & APEX) **Date: 10-01-2025**
Time: 3 Hrs **GTM-16** **Max. Marks: 300**

PHYSICS

1	**A**	2	**A**	3	**A**	4	**D**	5	**D**
6	**D**	7	**C**	8	**A**	9	**B**	10	**D**
11	**A**	12	**A**	13	**C**	14	**C**	15	**A**
16	**A**	17	**C**	18	**B**	19	**D**	20	**A**
21	**1234**	22	**12**	23	**1234**	24	**124**	25	**124**

CHEMISTRY

26	**B**	27	**A**	28	**D**	29	**A**	30	**C**
31	**A**	32	**C**	33	**D**	34	**B**	35	**D**
36	**B**	37	**C**	38	**B**	39	**D**	40	**C**
41	**C**	42	**D**	43	**D**	44	**B**	45	**B**
46	**1**	47	**6**	48	**5**	49	**4**	50	**3**

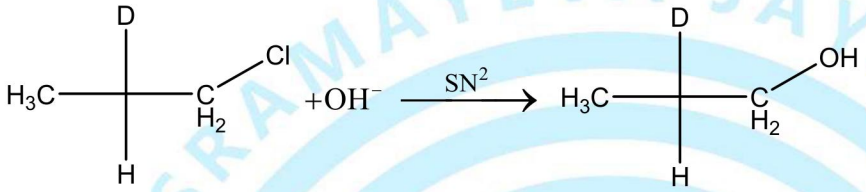
MATHS

51	**C**	52	**A**	53	**B**	54	**D**	55	**B**
56	**B**	57	**C**	58	**B**	59	**C**	60	**B**
61	**D**	62	**D**	63	**C**	64	**C**	65	**C**
66	**B**	67	**C**	68	**B**	69	**C**	70	**D**
71	**99**	72	**3**	73	**7**	74	**9**	75	**1006**

CHEMISTRY

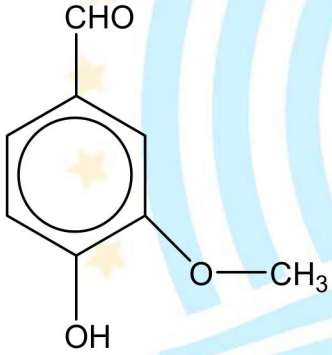
26. O_2F_2 is used to remove Pu
27. Hydronium ion further hydrated to positive ions.
28. Niacin, phenolphthalein and bromothymol blue are weak acids.
29. carbonic acid(Ka_2) is least and oxalic acid (Ka_2) is high.

30. The concentration of complex increases by adding KSCN.
31. As concentration decreases extent of dissociation increases
32. Stronger attraction forces between them leads to negative deviation
33. Conductivity of 0.1M KCl less than 0.1M HCl
34. In lead storage cell 38% H_2SO_4 used as electrolyte.
35. S,C, Ca & Al are in Mische metal.
36.

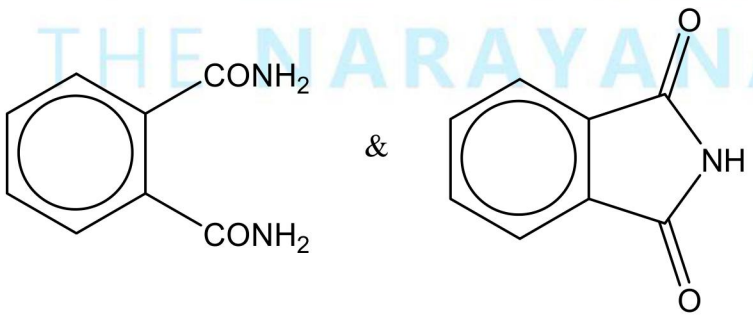


(R) (R)

37. Ka – order of alcohol is $i^o > 2^o > 3^o$
& pKa – order is $i^o < 2^o < 3^o$
38. Vanillin structure contains – aldehyde, Ether and phenolic groups.

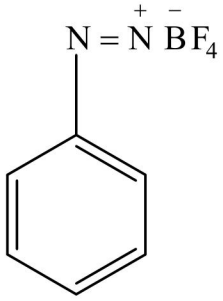


39. Benzoic acid is nearly insoluble in cold water.
40.



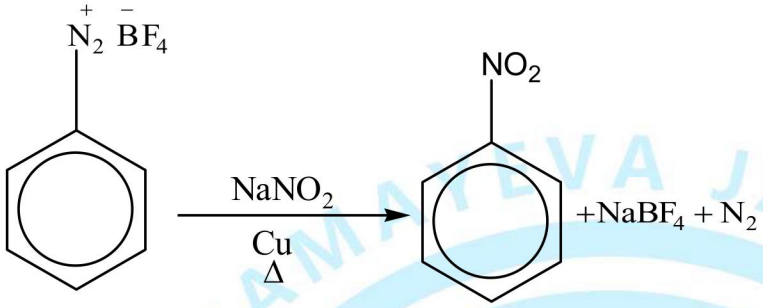
(Phthalamide) (Phthalimide)

41. Benadryl ($\overset{0}{3}$ - amine) , Adrenalilne ($\overset{0}{2}$ - amine)
42.



is a precipitate

43.



(Nitrobenzene)

44. Glucose does not reacts with $NaHSO_3$
45. α – Helix – Right handed screw.
46. X = 0.01, Y = 0.01
47. interstitial compounds have fractional oxidation states.
48. Pu & Np both exhibits +7 oxidation state.
49. Except H_3PO_4 all species undergo disproportionation(6)
50. Statements 2,3,4 are wrong.

MATHEMATICS

Let JP be the 1st given line and VS be the 2nd given line. The equation of the given line JP is

$$\frac{x-4}{1}=\frac{y+1}{-2}=\frac{z}{1} \quad \ldots (1)$$

Where $(4,-1,0)$ is a point and $1,-2,1$ are direction ratios of the given line JP. The equation of the given line VS is $\frac{9x-16}{13}=\frac{9y-1}{7}=\frac{z}{-1}$ i.e., $\frac{x-\frac{16}{9}}{\frac{13}{9}}=\frac{y-\frac{1}{9}}{\frac{7}{9}}=\frac{z}{-1}$

$$\Rightarrow \frac{x-\frac{16}{9}}{13}=\frac{y-\frac{1}{9}}{7}=\frac{z}{-9} \quad \ldots\ldots\ldots\ldots (2)$$

Where $\left(\frac{16}{9},\frac{1}{9},0\right)$ is a point and $13,7,-9$ are direction ratios of the line VS.

We know that the lines $\frac{x-x_P}{\ell_1}=\frac{y-y_P}{m_1}=\frac{z-z_P}{n_1}$ and

$\frac{x-x_V}{\ell_2}=\frac{y-y_V}{m_2}=\frac{z-z_V}{n_2}$ are coplanar then

$$\begin{vmatrix} x_P - x_V & y_P - y_V & z_P - z_V \\ \ell_1 & m_1 & n_1 \\ \ell_2 & m_2 & n_2 \end{vmatrix} = 0 \quad .....(3)$$

Here $\ell_1 = 1, m_1 = -2, n_1 = 1, \ell_2 = 13, m_2 = 7, n_2 = -9$

$x_P = 4, y_P = -1, z_P = 0, x_V = \frac{16}{9}, y_V = \frac{1}{9}, z_V = 0$

Putting these values in (3), we get

$$\therefore \begin{vmatrix} 4 - \frac{16}{9} & -1 - \frac{1}{9} & 0 - 0 \\ 1 & -2 & 1 \\ 13 & 7 & -9 \end{vmatrix} = 0 \Rightarrow \begin{vmatrix} \frac{20}{9} & \frac{-10}{9} & 0 \\ 1 & -2 & 1 \\ 13 & 7 & -9 \end{vmatrix} = 0$$

$$\Rightarrow \frac{10}{9}\begin{vmatrix} 2 & -1 & 0 \\ 1 & -2 & 1 \\ 13 & 7 & -9 \end{vmatrix} = 0$$

$$\Rightarrow \frac{10}{9}\left[2\begin{vmatrix} -2 & 1 \\ 7 & -9 \end{vmatrix} + 1\begin{vmatrix} 1 & 1 \\ 13 & -9 \end{vmatrix}\right] = 0$$

$\Rightarrow 2(18-7)+1(-9-13) = 0$

$\Rightarrow 22 - 22 = 0$ Which is true.

Hence the assertion is true.

Reason: From equation (3)

Reason is wrong.

52. Let OJ be a line whose direction cosines are ℓ_1, m_1, n_1

Let OP be a line whose direction cosines ae ℓ_2, m_2, n_2

Let OV be a line whose direction cosines are ℓ_3, m_3, n_3

Let OS be a line whose direction cosines are

$$\frac{\ell_1 + \ell_2 + \ell_3}{\sqrt{3}}, \frac{m_1 + m_2 + m_3}{\sqrt{3}}, \frac{n_1 + n_2 + n_3}{\sqrt{3}}$$

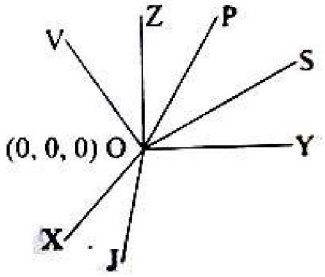
Let α be the angle between OS and OJ then

$$\cos\alpha = \frac{\ell_1(\ell_1 + \ell_2 + \ell_3)}{\sqrt{3}} + \frac{m_1(m_1 + m_2 + m_3)}{\sqrt{3}} + \frac{n_1(n_1 + n_2 + n_3)}{\sqrt{3}}$$

$$\ell_1^2 + m_1^2 + n_1^2 + \ell_1\ell_2 + m_1m_2 +$$

$$= \frac{n_1n_2 + \ell_1\ell_3 + m_1m_3 + n_1n_3}{\sqrt{3}}$$

$\therefore OJ, OP, OV$ are mutually perpendicular lines then



$\ell_1\ell_2 + m_1m_2 + n_1n_2 = 0$

$\ell_1\ell_3 + m_1m_3 + n_1n_3 = 0$

$\ell_2\ell_3 + m_2m_3 + n_2n_3 = 0$

$\therefore \ell_1^2 + m_1^2 + n_1^2 = 1$

$\therefore \ell_2^2 + m_2^2 + n_2^2 = 1$

$\therefore \ell_3^2 + m_3^2 + n_3^2 = 1$

$\therefore \cos\alpha = \frac{1}{\sqrt{3}}$

Let β be the angle between OS and OP. γ be the angle between OS and OV.

Simplify $\cos\beta = \frac{1}{\sqrt{3}}, \cos\gamma = \frac{1}{\sqrt{3}}$

Hence the assertion is true.

Obviously reason is true

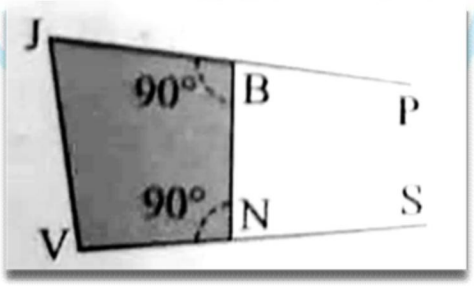
53. Assertion: Let JP be the first given line and VS be the second line. The equation of the given line JP is $\frac{x+3}{-4} = \frac{y-6}{3} = \frac{z}{2} = \lambda(\text{say})$

Where $J \equiv (-3,6,0)$ is a point and $-4,3,2$ are direction ratios of JP.

Let VS be the second line Equation of 2nd line VS is $\frac{x+2}{-4} = \frac{y}{1} = \frac{z-7}{1} = \mu$ (say).

Where $V \equiv (-2,0,7)$ is a point and $-4,1,1$ are direction ratios of line VS.

If BN is the shortest distance between JP and VS then $BN \perp JP$ and $BN \perp VS$.



Let ℓ, m, n be the direction ratios of the line BN. If $BN \perp JP$ then

$-4\ell + 3m + 2n = 0$ .....(1)

If $BN \perp VS$ then $-4\ell + m + n = 0$ ....(2)

From (1) and (2)

$-4\ell - m - n = 0$

$$\frac{-4\ell-3m-2n=0}{\frac{\ell}{2-3}=\frac{m}{-4+8}=\frac{n}{-12+4}}$$ by cross multiplication

$$\Rightarrow \frac{\ell}{-1}=\frac{m}{4}=\frac{n}{-8}=\frac{\sqrt{\ell^2+m^2+n^2}}{\sqrt{1+16+64}}=\frac{1}{9}$$

$$\therefore \ell=-\frac{1}{9}, m=\frac{4}{9}, n=-\frac{8}{9}$$

$\therefore$ The shortest distance = BN

$$=\left|(-2+3)\left(-\frac{1}{9}\right)+(0-6)\frac{4}{9}+(--0)\left(-\frac{8}{9}\right)\right|$$

$$=\left|-\frac{1}{9}-\frac{24}{9}-\frac{56}{9}\right|=\left|\frac{81}{9}\right|=9$$

Reason: We know that if two lines are skew lines or non intersecting lines then they do not lie on the same plane.

54. $f(x)=\frac{1}{x^2+2(a-3)x+(a-1)}$ is continuous for disc. <0

$$\Rightarrow 4(a-3)^2-4(a-1)<0$$

$$\Rightarrow a^2-6a+9-a+1<0$$

$$\Rightarrow a^2-7a+10<0 \Rightarrow (a-2)(a-5)<0$$

$$\Rightarrow a\in(2,5)$$

$\therefore$ for $2<a<5, \frac{1}{x^2+2(a-3)x+(a-1)}$ is continuous, and hence integrable

55. $A: I=\int_{-\pi}^{\pi}\frac{2x\left(1+\sin^2 x\right)}{1+\cos^2 x}dx$

$I=0 \qquad (\because f(x)=-f(-x))$

$\Rightarrow$ Assertion is true.

R: Let $I_1=\int_0^{\pi/2}\sin^2 x\cos^2 x(\sin x+\cos x)dx$ and $\sin x-\cos x=t$

$$\Rightarrow 1-2\sin x\cos x=t^2 \text{ and } (\cos x+\sin x)dx=dt$$

$$\Rightarrow \frac{1-t^2}{2}=\sin x\cos x$$

$$\therefore I_1=\int_{-1}^{1}\left(\frac{1-t^2}{2}\right)^2 dt$$

$$=\frac{1}{2}\int_0^1\left(1-2t^2+t^4\right)dt=\frac{1}{2}\left(t-\frac{2t^3}{3}+\frac{t^3}{5}\right)_0^1$$

$$=\frac{1}{2}\left[\frac{15t-10t^3+3t^5}{15}\right]_0^1=\frac{1}{2}\left(\frac{15-10+3}{15}\right)=\frac{4}{15}$$

$\Rightarrow$ Reason is also true, but there is no connectivity between the assertion and reason.

56. Given differential equation is

$$\frac{y+\sin x\cos^2(xy)}{\cos^2(xy)}dx+\left(\frac{x}{\cos^2(xy)}+\sin y\right)dy=0$$

$$\Rightarrow\left(y\sec^2(xy)+\sin x\right)dx+\left(x\sec^2(xy)+\sin y\right)dy=0$$

$$\Rightarrow\sec^2(xy)(xdy+ydx)+\sin xdx+\sin ydy=0$$

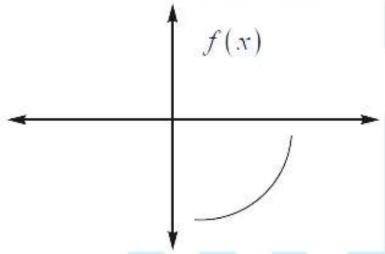
$$\Rightarrow d(\tan xy)+d(-\cos x)+d(-\cos y)=0$$

$$\Rightarrow\tan(xy)=\cos x+\cos y+C$$

59. Conceptual

60. Statement-II: $f(x)=-x^3$ at $x=0$ contradiction

Statement-III: $\left(f^2(x)\right)^1=2f(x)f'(x)=-ve$



Statement-IV: Contradiction

61. Let $f:R\to R$ is defined by $f(x)=\left(x^2+\sin x\right)(x-1)$

Then, $f\left(1^+\right)=f\left(1^-\right)=f(1)=0$

Let $(fg):R\to R$ is defined by $(fg)(x)=f(x).g(x)$

Let $(fg)=h(x)=f(x).g(x)$ then $h:R\to R$

$h'(x)=f'(x)g(x)+f(x)g'(x)$

If g is differentiable at x = 1

$h'(1)=f'(1)g(1)+0,\ [\because f(1)=0]$

If $g(x)$ is differentiable then $h(x)$ I also differentiable (true)

If $g(x)$ is differentiable at x = 1, then *fg* is also differentiable at x = 1

If g(c) is continuous at x= 1, then $g\left(1^+\right)=g\left(1^{-1}\right)=g(1)$

$$h'\left(1^+\right)=\lim_{h\to0^+}\frac{h(1+h)-h(1)}{h}$$

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$$h'(1^+)=\lim_{h\to 0^+}\frac{f(1+h)g(1+h)-0}{h}=f'(1)g(1)$$

$$h'(1^-)=\lim_{h\to 0^+}\frac{f(1-h)g(1-h)-0}{-h}=f'(1)g(1)$$

$\Rightarrow h(x)=f(x).g(x)$ is differentiable at x = 1 (True)

So, if g is continuous at x = 1, THEN *fg* is differentiable at x =1.

Option (b) (d) $h'(1^+)=\lim_{h\to 0^+}\frac{h(1+h)-h(1)}{-h}$

$$h'(1^+)=\lim_{h\to 0^+}\frac{f(1+h)g(1+h)}{h}=f'(1)g(1^+)$$

$$h'(1^-)=\lim_{h\to 0^+}\frac{f(1-h)g(1-h)}{-h}=f'(1)g(1^-)$$

$\Rightarrow g(1^+)=g(1^-)$

So, it does not mean that if *fg* is differentiable at x= 1, then *fg* is continuous or differentiable at x = 1

62. Let $A=\{0,1,2,3,4,5,6,7\}$

To find the number of bijective functions $f:A\to A$ such that $f(1)+f(2)+f(3)=3$, the only possibility for this sum is 0 + 1 + 2 =3.

$\Rightarrow$ Elements 1, 2, 3 in the domain can be mapped with 0, 1, 2 only.

Therefore, number of bijective functions = 3! ×5! = 6 × 120 = 720.

63. $AB=0|B|\neq 0\ \ AB=0\Rightarrow A=0$

64. $$\Delta=\begin{vmatrix}-1 & 1 & 2\\ 3 & -a & 5\\ 2 & -2 & -a\end{vmatrix}$$

$$=-1(a^2+10)-1(-3a-10)+2(-6+2a)$$

$$=-a^2-10+3a+10-12+4a$$

$$\Delta=-a^2+7a-12$$

$$\Delta=-\left[a^2-7a+12\right]$$

$$\Delta=-\left[(a-3)(a-4)\right]$$

$$\Delta_1=\begin{vmatrix}0 & 1 & 2\\ 1 & -a & 5\\ 7 & -2 & -a\end{vmatrix}=0-1(-a-35)+2(-2+7a)$$

$\Rightarrow a+35-4+14a$

$15a+31$

Now $\Delta_1=15a+31$

For inconsistent $\Delta = 0$

$\therefore$ a=3, a = 4 and for a = 3 and 4

$\Delta_1 \neq 0$

$n(S_1) = 2$

For infinite solution: $\Delta = 0$ and $\Delta_1 = \Delta_2 = \Delta_3 = 0$

Not possible

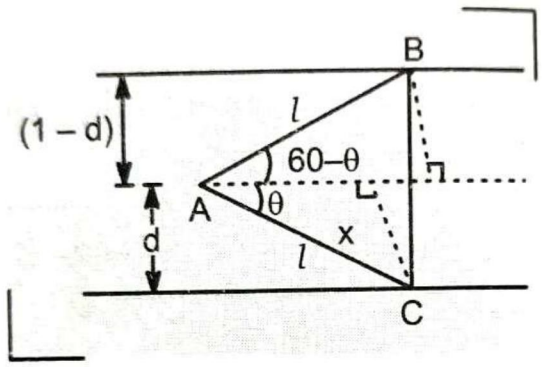
$\therefore n(S_2) = 0$

65. Let $\vec{a} \wedge \vec{b} = \theta$, Now $\vec{x} = \vec{a} - (\vec{a}.\vec{b})\vec{b}$ $\Rightarrow |\vec{x}|^2 = \vec{a}^2 + (\vec{a}.\vec{b})^2 b^2 - 2(\vec{a}.\vec{b})^2$

$\Rightarrow |\vec{x}|^2 = 1 + \cos^2\theta - 2\cos^2\theta$ $\Rightarrow |\vec{x}| = \sin\theta, Also \therefore \vec{y} = \vec{a} \times \vec{b} \Rightarrow |\vec{y}| = |\vec{a} \times \vec{b}| = \sin\theta$

$\therefore |\vec{x}| = |\vec{y}|$ Also, $\vec{x}.\vec{b} = \vec{a}.\vec{b} - (\vec{a}.\vec{b})\vec{b}^2 = 0$ $\therefore |\vec{x}.\vec{b}| = 0$

66.



$l\sin\theta = d$ ...(1)

$l\sin(60-\theta) = l\frac{\sqrt{3}}{2}\cos\theta - \frac{l}{2}\sin\theta$

$= l - d$

$\Rightarrow l\cos\theta = \frac{2-d}{\sqrt{3}}$ .... (2)

Squaring and adding Equations. (1) and (2)

$\Rightarrow l^2 = d^2 + \frac{4 + d^2 - 4d}{33} = \frac{4}{3}(d^2 - d + 1)$

$\Rightarrow l = 2\sqrt{\frac{d^2 - d + 1}{3}}$

68. $I.F = e^{\int \frac{xdx}{x^2-1}} = \sqrt{1-x^2}$

$y = f(x) = \frac{3x^5 + x^4 + 2x^2 + x}{\sqrt{1-x^2}} = \int_{-1/2}^{1/2} \frac{f(x)}{(x^2+2)} dx = \frac{\pi}{6} - \frac{1}{4}$

69. $$S_k=\sum_{r=1}^{k}\tan^{-1}\left(\frac{\frac{1}{3}\left(\frac{2}{3}\right)^r}{\left(\frac{2}{3}\right)^{2r+1}+1}\right)=\sum_{r=1}^{k}\left[\tan^{-1}\left(\frac{2}{3}\right)^r-\tan^{-1}\left(\frac{2}{3}\right)^{r+1}\right]$$

$$=\tan^{-1}\left(\frac{2}{3}\right)-\tan^{-1}(0)=\cot^{-1}\left(\frac{3}{2}\right)$$

70. $$2|=2\int_{\pi/8}^{3\pi/8}\frac{11+\cos 4x}{1-\cos 4x}dx$$

$$\Rightarrow|=|=\int_{\pi/8}^{3\pi/8}\frac{12-(1-\cos 4x)}{1-\cos 4x}=12\int_{\pi/8}^{3\pi/8}\frac{1}{2\sin^2 2x}dx-\int_{\pi/8}^{3\pi/8}dx=6\int_{\pi/8}^{3\pi/8}\cos ec^2 2xdx-\frac{\pi}{4}$$

$$=\left[-\frac{6}{2}\cot 2x\right]_{\frac{\pi}{8}}^{\frac{3\pi}{8}}-\frac{\pi}{4}=-3\left[(-1)+(-1)\right]-\frac{\pi}{4}=6-\frac{\pi}{4}$$

71. $$f(1,m+1)=f(1,m)+f(m,m+1)-2f(1,m)f(m,m+1)$$

$$=f(1,m)+\frac{1}{3}-\frac{2}{3}f(1,m)=\frac{1}{3}+\frac{1}{3}f(1,m)$$

Now

$$f(1,100)=\frac{1}{3}+\frac{1}{3}f(1,99)=\frac{1}{3}+\frac{1}{3}\left(\frac{1}{3}+\frac{1}{3}f(1,98)\right)$$

$$=\frac{1}{3}+\frac{1}{3^2}+\frac{1}{3^2}f(1,98)=\frac{1}{3}+\frac{1}{3^2}+\frac{1}{3^3}+\frac{1}{3^3}f(1,97)$$

$$=\frac{1}{3}+\frac{1}{3^2}+\frac{1}{3^3}+.....+\frac{1}{3^{98}}+\frac{1}{3^{98}}f(1,2)$$

$$=\frac{1}{3}+\frac{1}{3^2}+\frac{1}{3^3}+.....+\frac{1}{3^{98}}+\frac{1}{3^{99}}=\frac{1}{2}\left(1-\frac{1}{3^{99}}\right)$$

72. $$(Sinx+3)\left(\sin x-(x-1)^2\right)=0$$

73. $$\text{Required}=\lim_{n\to\infty}\frac{3\left(\sum_{r=1}^{n-1}\left(-r^3+r^2(n+1)-nr\right)\right)}{n(n+1)(3n^2-n-2)}=\frac{1}{6}$$

74. We know that y > 0 and $y=\frac{10}{\sqrt{x}}$.

$$(\log x)^2+(\log y)^2=(\log x)^2+\left(\log\frac{10}{\sqrt{x}}\right)^2=(\log x)^2+\left(1-\frac{1}{2}\log x\right)^2$$

$$=\frac{5}{4}(\log x)^2-\log x+1=\frac{5}{4}\left(\log x-\frac{2}{5}\right)^2+\frac{4}{5}$$

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Since $1 \le x \le 10$, then $0 \le \log x \le 1$

When $\log x = \frac{2}{5}$, or $x = 10^{\frac{2}{5}}, (\log x)^2 + (\log y)^2$ has the smallest value of $\frac{4}{5}$. The answer is $4 + 5 = 9$.

75. For example, there are $2 \times 1 + 1 = 3$ integers with the value of 1.

$\left[\sqrt{1}\right] = 1, \left[\sqrt{2}\right] \approx [1,414] = 1, \left[\sqrt{3}\right] \approx [1.732] = 1,$

There are $2 \times 2 + 1 = 5$ integers with value of 2

$\left[\sqrt{4}\right] = 2, \left[\sqrt{5}\right] \approx [2.236] = 2, \left[\sqrt{6}\right] \approx [2.449] = 2, \left[\sqrt{7}\right] = 2, \left[\sqrt{8}\right] = 2,$

$2016 > 1936 = 44^2$

$2016 - 1936 + 1 = 81$

So the sum will be $1 \times 3 + 2 \times 5 + ... + 43 \times 87 + 44 \times 81$

$= \sum_{n=1}^{43} n(2n+1) + 44 \times 81$

$= \frac{2}{6} \times 43 \times 44 \times 87 + \frac{1}{2} \times 43 + 44 \times 81$

= 59378

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